

Auteur: Shahin Jamal
Studierichting: Informatica
Oefeningenreeks 2D nr. 10.19

$$\begin{aligned} & \lim_{x \rightarrow 3} \frac{x-3}{\sqrt{x^2-9}} \\ & \lim_{x \rightarrow 3+} \frac{x-3}{\sqrt{x^2-9}} \cdot \frac{\sqrt{x^2-9}}{\sqrt{x^2-9}} \\ & = \lim_{x \rightarrow 3+} \frac{(x-3)(\sqrt{x^2-9})}{x^2-9} \\ & = \lim_{x \rightarrow 3+} \frac{(x-3)\sqrt{x^2-9}}{(x-3)(x+3)} = 0 \\ & \lim_{x \rightarrow 3-} \frac{x-3}{\sqrt{x^2-9}} \text{ bestaat niet.} \end{aligned}$$

References